

Secondhand Marketplace2

Beta Viva

Pre-GreenCheck AI

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3/3/2026

1

Now speaking: Euan

Journey from previous project to now

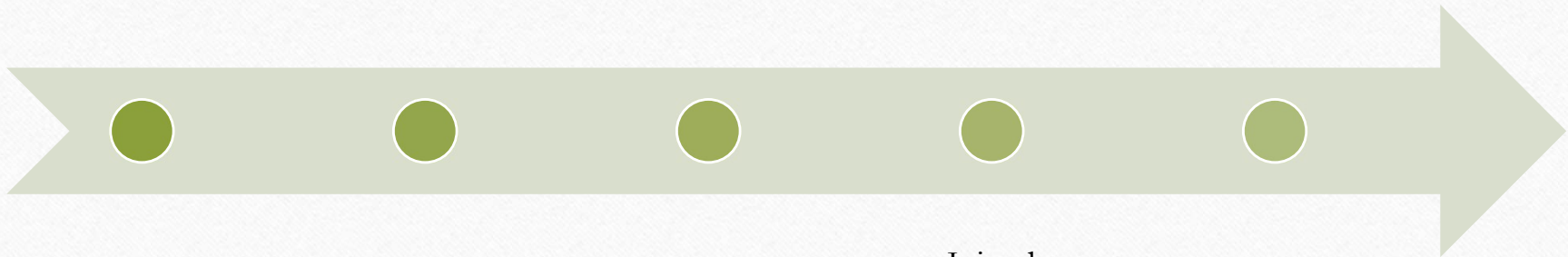
Joined
GreenCheck
AI

Deciding and
choosing new
project

Implementation
in progress

Find out issues
and risks

Joined
Secondhand
Marketplace in
week 7



Previous Project: GreenCheck AI

Greenwashing detection using XAI for Legal Pythia

Key Issues

- 100% of frontend and >50% of backend AI-generated
- CTO unable to clearly explain the codebase, “XAI” wrapper around Gemini
- Past student work misrepresented as commercial success

You are an expert ESG analyst providing transparent, explainable assessments with source attribution.

IMPORTANT: You MUST respond ENTIRELY in ENGLISH. All explanations, quotes, summaries and recommendations must be in English.

CRITICAL: For EVERY score you assign, you MUST provide:

1. The exact reasoning why you chose that specific number
2. Which specific information from the sources influenced the score
3. What information was missing that prevented a higher score
4. Direct quotes from sources when available (translated to English if necessary)

Previous Project: GreenCheck AI

Greenwashing detection using XAI for Legal Pythia

The Pivot

- Discontinued the existing codebase following Sarah's advice
- Pivoted to the Secondhand Marketplace

Team Update

- One member departed in TB1; one joined in TB2
- Final team size: 4

Now speaking: Jen

Current Project *Context* & Vision

Marius Jurt is an individual client
who aims to address the lack of
trust and engagement in existing
secondhand marketplaces

Now speaking: Jen

Current Project Context & *Vision*

To build a trusted, playful marketplace where users trade
confidently and build credibility

User Stories

Buyer

Seek an engaging and interactive second-hand shopping experience with item recommendations and access to seller reviews

Partially implemented: Marketplace UI & Explore page (Tinder-style suggestions), Liked Items list

Seller

Require a straightforward method to list items and reach the appropriate audience, encouraging community participation

Partially implemented: Selling items functionality; My Listings page planned

Stakeholders



Viewer



**Postal Service
Providers**



**Payment Service
Providers**

Now speaking: Freddie

DEMO

Project Architecture

Frontend

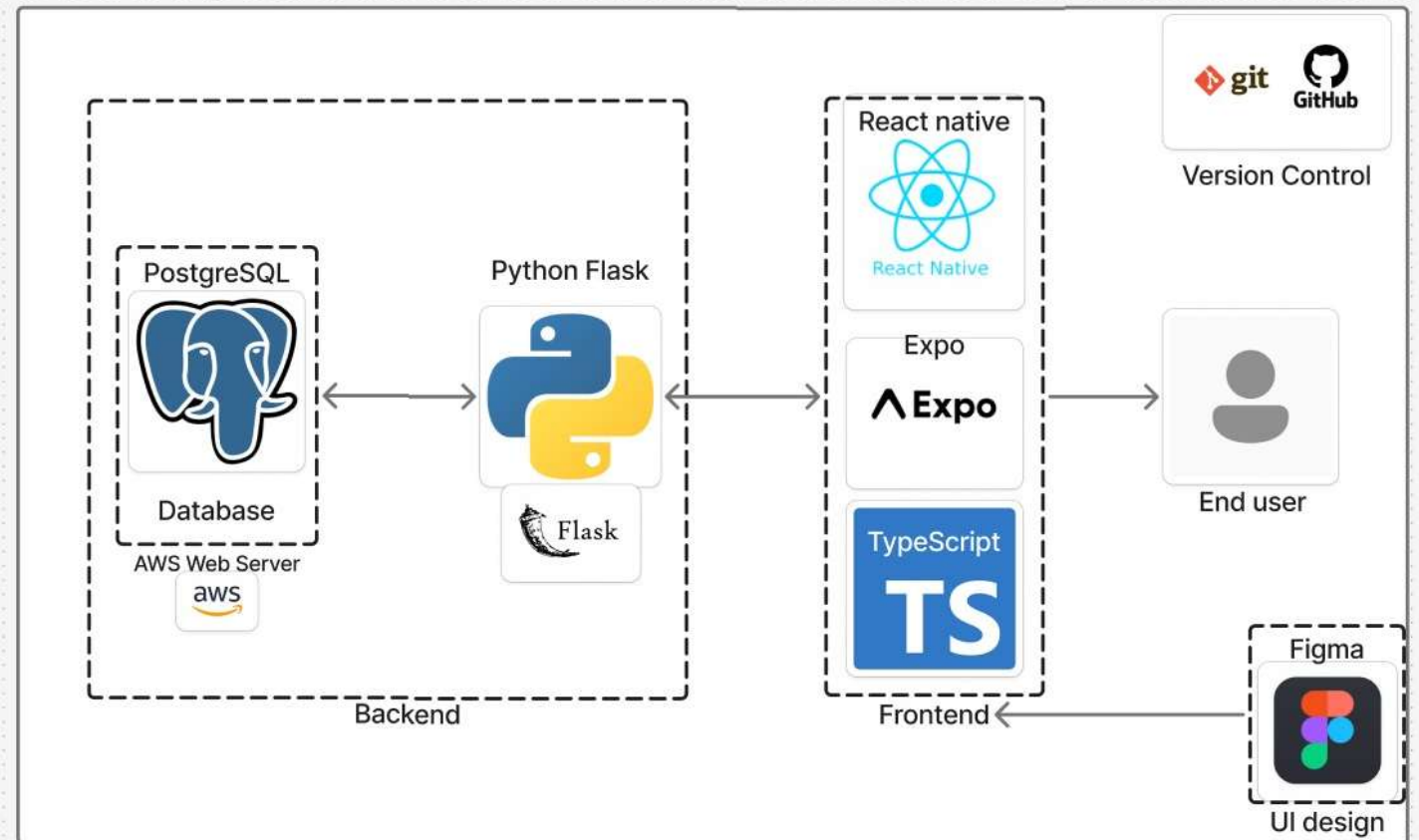
- React Native – Expo

Backend

- PostgreSQL Database
- Flask & Python

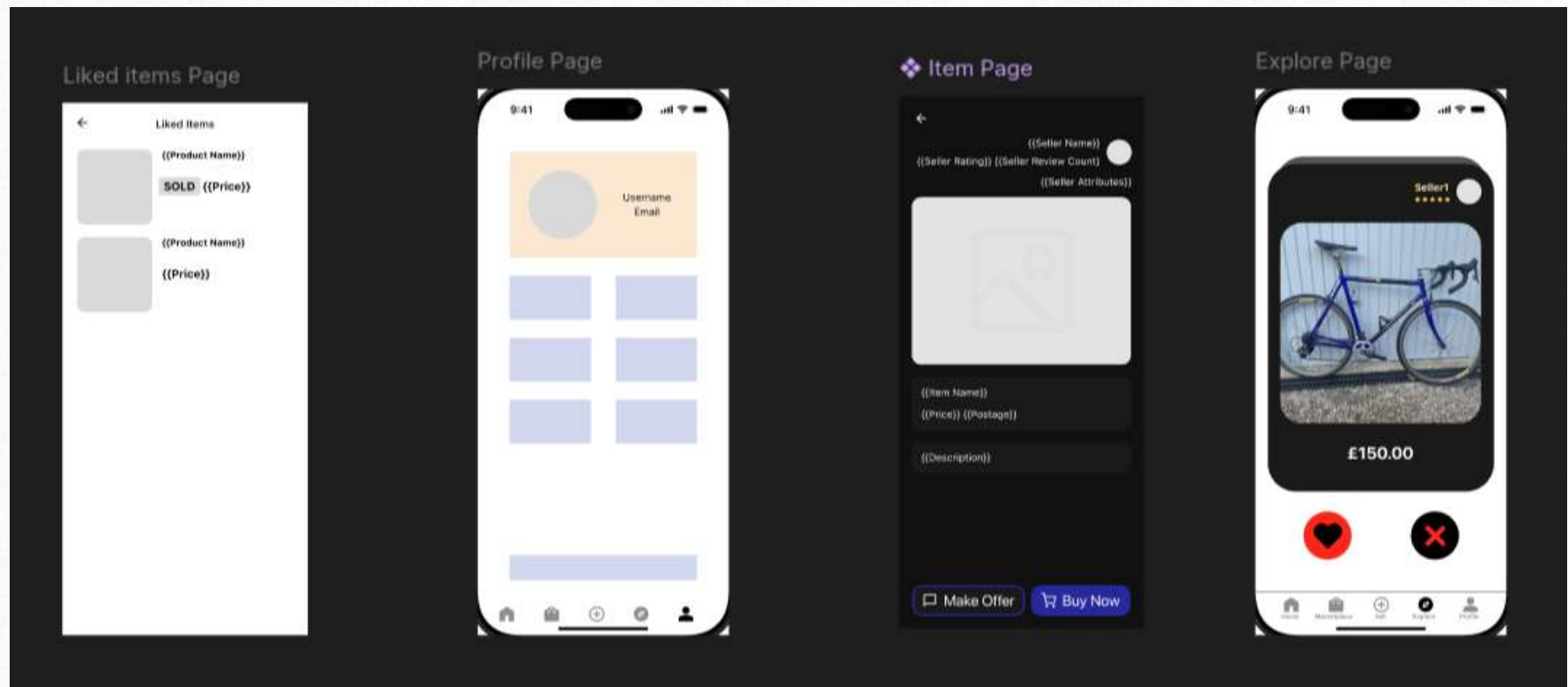
Version Control & Prototyping

- Figma
- Github & Git

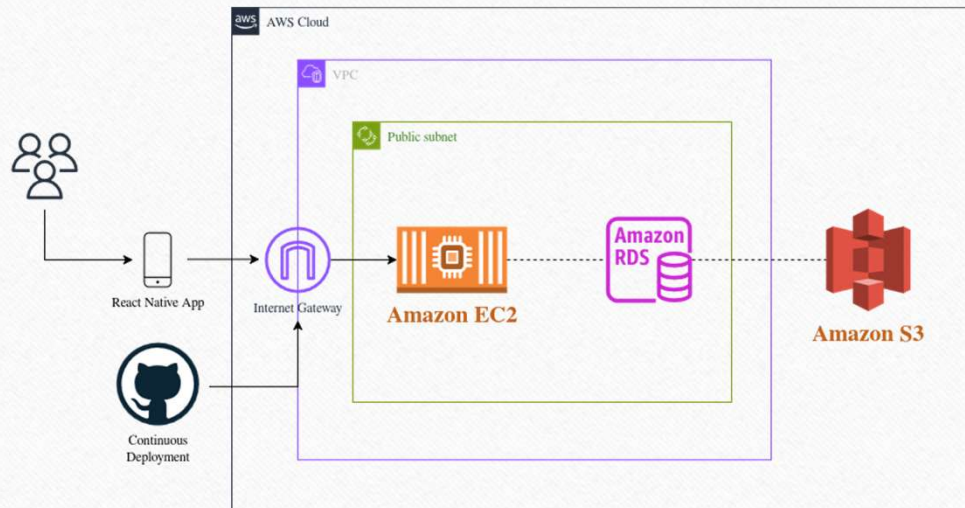


Now speaking: Alex

Figma Designs



Cloud Architecture



Amazon EC2 – Backend

- Our backend server runs on an Amazon EC2 instance using a Flask application served by Gunicorn
- Nginx securely handles all incoming HTTP traffic and forwards them to our backend

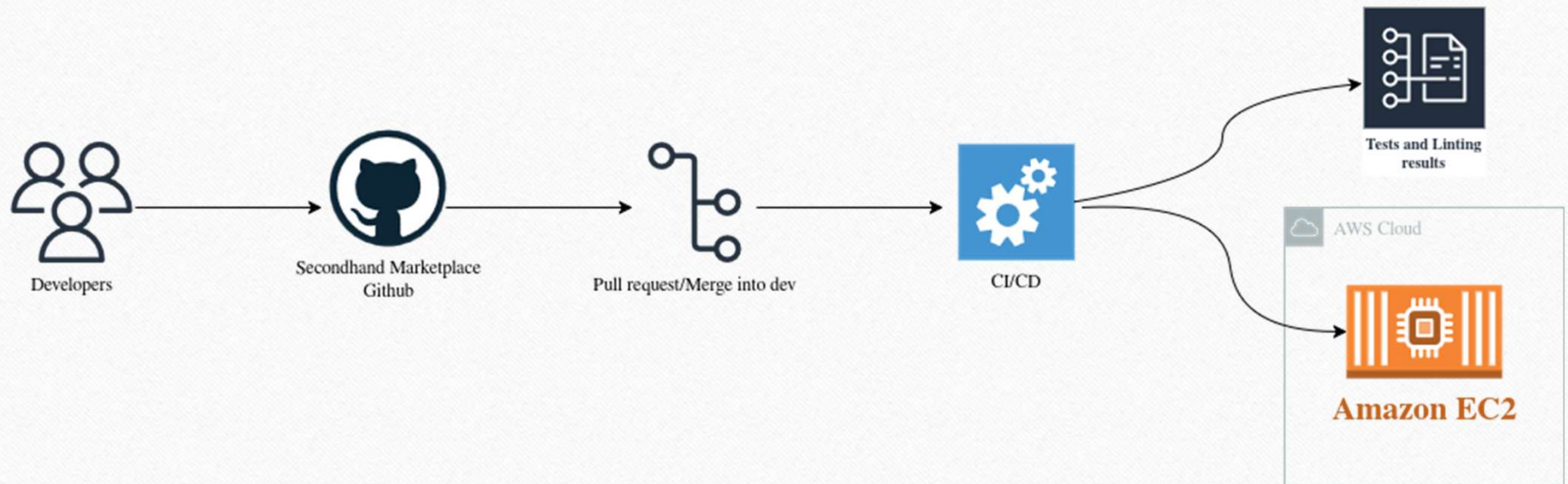
Amazon RDS – Database

- We use an Amazon RDS instance to host our PostgreSQL database

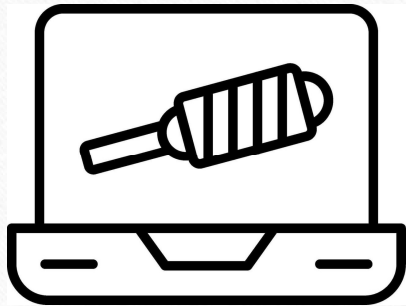
Amazon S3 – Cloud Storage

- Uploaded product images from the user are processed by the backend and sent to an Amazon S3 bucket

Continuous Integration and Deployment

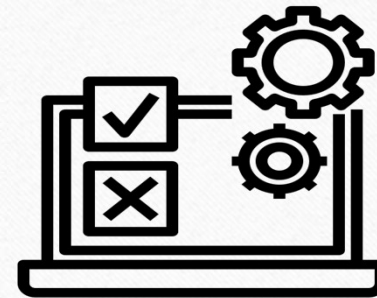


Continuous Integration



Linting Scripts

Ensure python code is consistent



Testing

Run tests on the frontend

Continuous Deployment



On a push to dev the workflow does the following:

- SSH into the EC2 instance
- Pull the latest code from dev
- Restart the flask server

Next steps:

- Have two flask servers running on the EC2 instance, a main one that updates when merging into dev, and a testing one that updates when a pull request is made so we can spot issues before merging into dev

Client feedback

Latest feedback:

- Generally happy and thinks we know what we are doing
- Propose new ideas for gamification
- Pointed out some areas of improvement

Client 19th Feb:

- Feedback:
 - Want swiping to be more reactive to the end - can keep it there and not disappear immediately
 - Liked about testing day feedback
 - Ask about the buy now and make offer button on item page: We need to decide to keep it or delete it, with a good reason and following user journey idea.
 - Happy with what we are having now, and think that we know what we are doing
- Idea proposal: Swiping creating a match if 2 users liked each other's item (Users who liked your items = User A)
 - Increase uniqueness: pure exchange of item, can pay difference in value or other ideas
 - Smart recommendation: increase users A items chance to appear in explore page
 - Up to us to decide if we want to develop on this and how to execute
- Butterflies: can use testathon to try different versions of butterfly interaction and see which is better?

User feedback of testing day - General trend

Method: Give user the app from home screen and let them explore it freely, then voice recorded interview is held.

Feedback of 10 total users:

- 80% complimented our UI and interface
- 50% complimented swiping feature
- Average 2.67/5 score on “how different from others”

General trend:

- Users liked:
 - 80% of users complimented our UI and interface:
 - clean, modern, minimalistic and easy to access.
 - 50% of users complimented our swiping feature:
 - allow users to discover, interactive and unique in online market
 - Easy to list things
- Users pointed out areas that can be improved / implemented:
 - Marketplace:
 - Labels for items, Searching can find related items & Sort by categories
 - Tapping on seller doesn't give details
 - Contact seller
 - Swiping
 - small description area
 - buy or not buy feature
 - Others:
 - When exiting page, on the top left is (tabs) not back
 - Empty home page
 - Liked items page
 - Prefer webpage
- Average Score of 2.67/5 on "How different do you think this app is compare to other online market"

User feedback of testing day - Liked items page

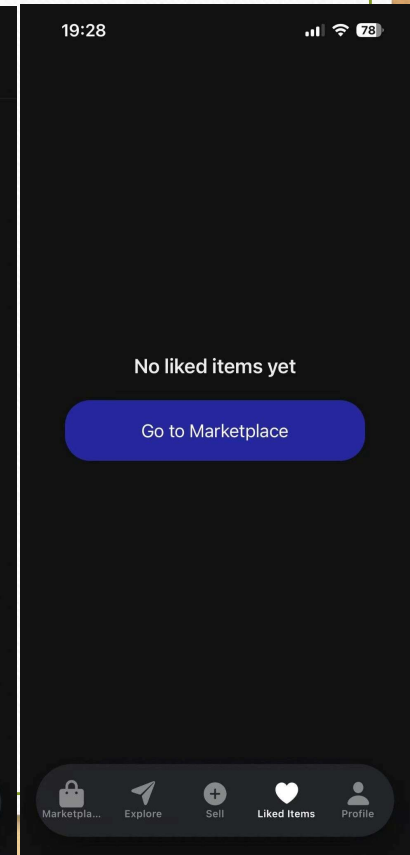
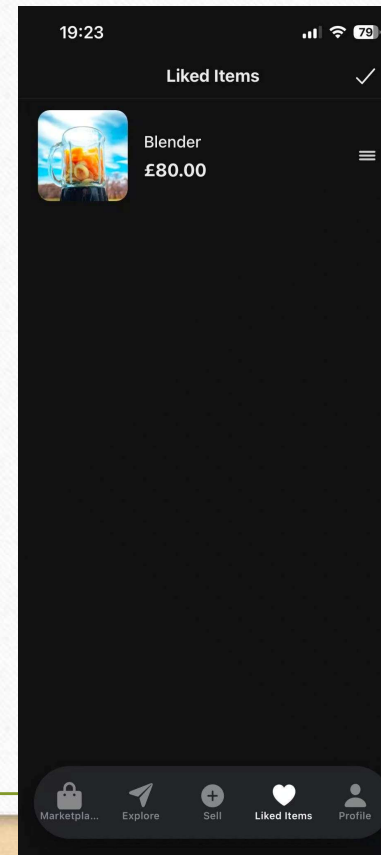
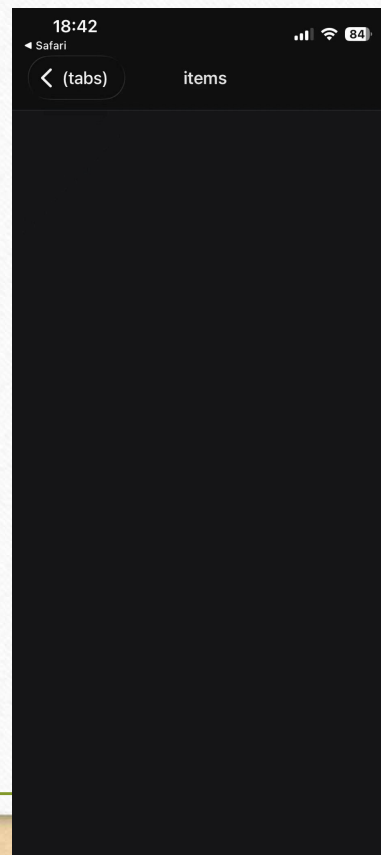
Before

After

User mentioned about should be able to have a liked items page to check the items they liked during swiping or exploring.

This enable them to check items they liked and increase chance of buying.

Currently frontend done, backend in progress.



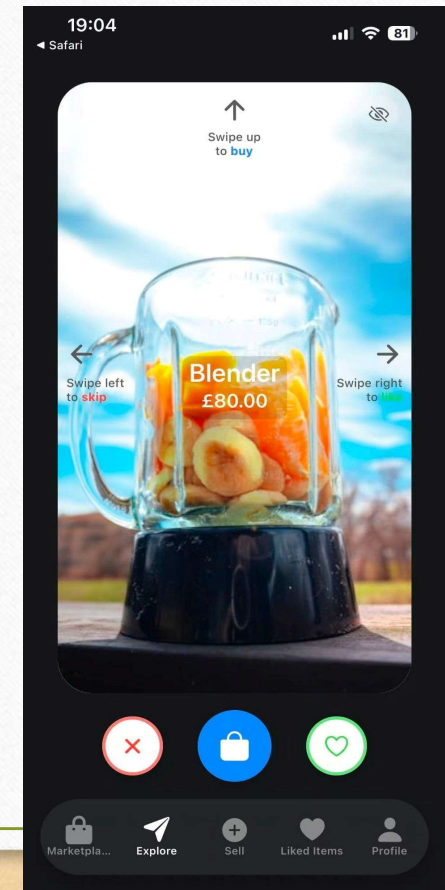
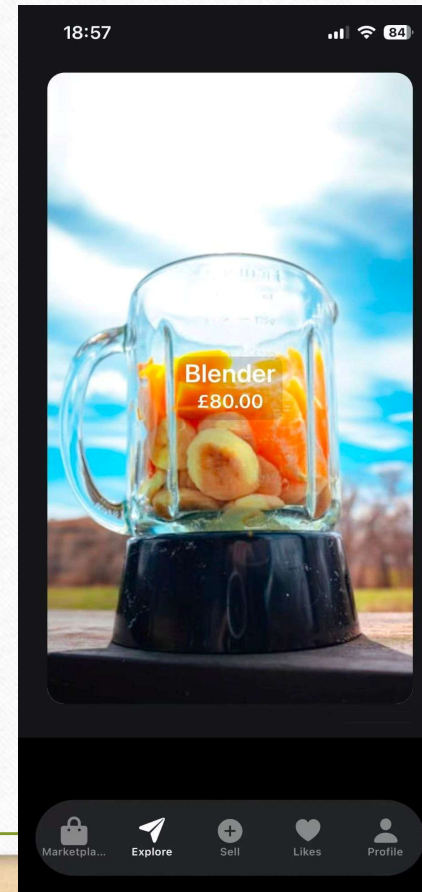
User feedback of testing day - Explore page enhancement

Before

After

User mentioned about having buy / skip feature for swiping. Some needed help for explaining what is this for.

We updated the buttons and text annotation for swiping. The buttons allow users to directly like or skip, while the text, can be hidden, provides instruction on how to use for swiping.



Now speaking: Jen

User feedback of testing day - Navigation improvement

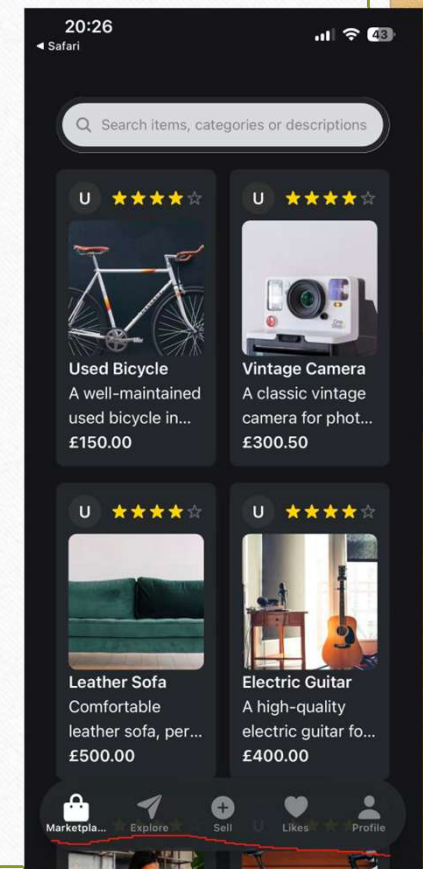
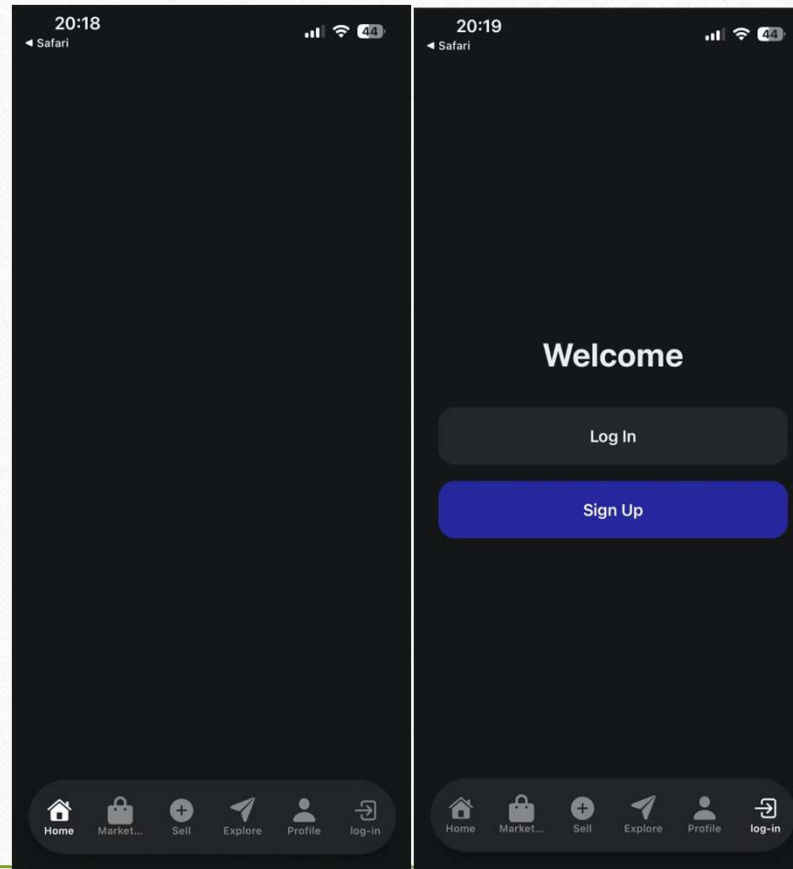
Before

After

Users mentioned about having an empty landing screen does not make sense.

Client also had feedback on this to let us simplify more.

Marketplace being the default screen with simplified navigation provides simpler and more direct use.

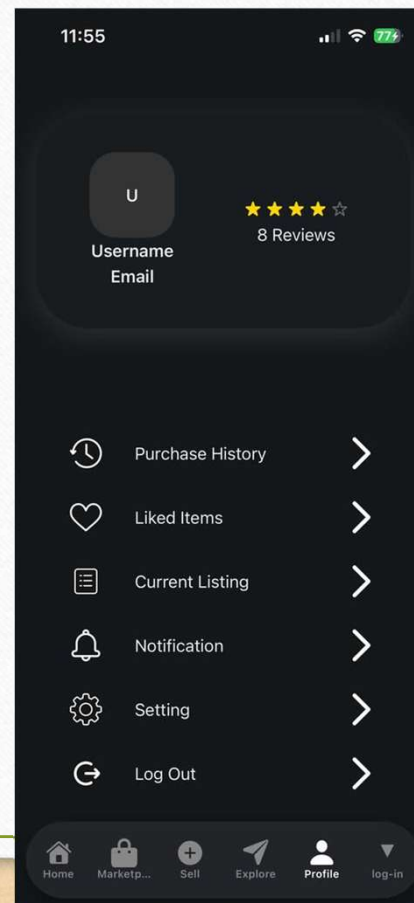


Client feedback - Profile page simplification

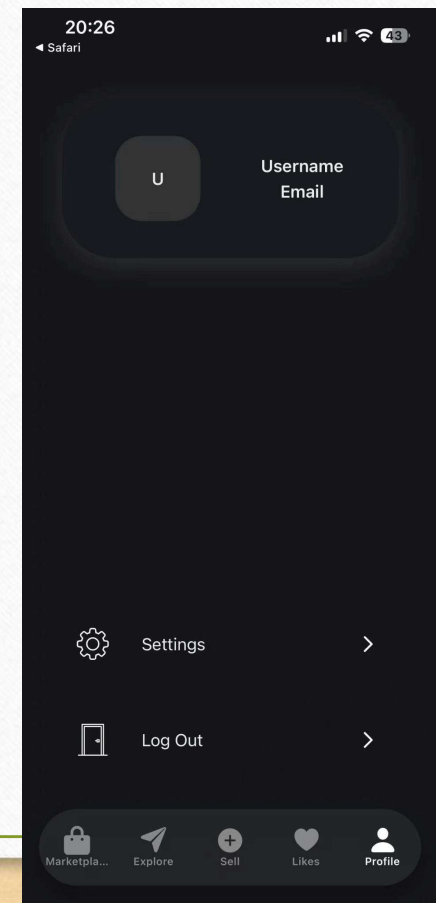
Client was concerned about profile page might be overloading for users.

We have simplified this page to be left with only needed functionalities and adjusted the reviews and row gap to make it remain minimal but functional.

Before



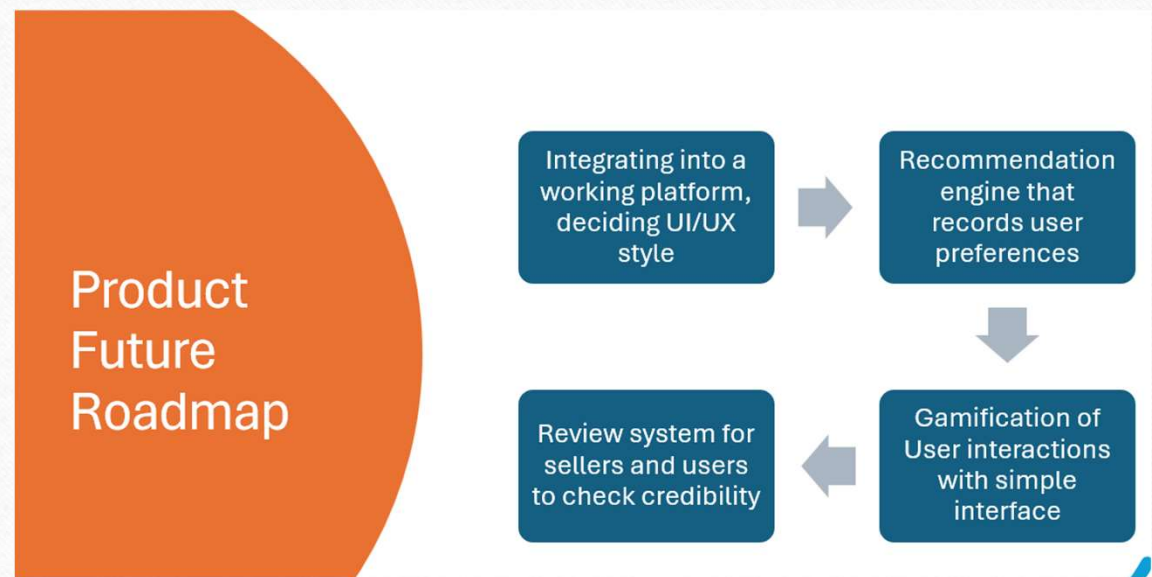
After



MVP feedback - Choose areas to be focused on

MVP feedback pointed out it is impossible to obtain all 4 areas.

We currently focus only on UI/UX and gamification and it is communicated with client to finalize this too. Roadmap has been adjusted accordingly.



MVP feedback - Effect of chosen gamification on app

MVP feedback pointed out we need to consider about effects of chosen design (haptics and swiping) on user and how does it affect the app.

We have done some research which explains the effect and also helped to provide ideas for future gamification enhancement for a marketing app.

Research findings on UX and gamification:

A comparison of the effects of haptic and visual feedback on presence in virtual reality

- Presence was enhanced when they could both see and feel the ball's action; with a strong suggestion that haptic feedback alone gave rise to a greater sense of presence than visual alone.
- Although its for VR, it can also be applied in mobile apps

Expectations for User Experience in HapticCommunication with Mobile Devices

- Subjective quality of haptic feedback: pleasant vibration, meaningful distinguishable patterns, realistic tactile sensations and customizable feedback
- Spontaneity & lightweight interaction: quick, effortless, immediate, yes/no signals
- Reassurance (of swiping actually does something) can be also considered

Analyzing Gamification of "Duolingo" with Focus on Its Course Structure

- Gamification is a term that refers to the use of game-based elements such as mechanics, aesthetics, and game thinking in non-game contexts aimed at engaging people, motivating action, enhancing learning and solving problem
- Game refinement theory
- The essence of gamification is that it occurs in a non-game context
- Making it more engaging and challenging for student

Swiping vs. Scrolling in Mobile Shopping Applications

- Suggests:
 - Swiping interface leads to greater cognitive absorption and playfulness in shopping applications - is significant in increasing reuse intentions and task performance.
 - Cognitive absorption: a state of deep involvement with software
 - Greater level of cognitive absorption, greater levels of engagement and enjoyment -> more accurate assessments of the streamed products & greater intention to reuse the app
 - Playfulness is associated with higher reuse intention BUT may reduce task performance -> saving products that are not of sufficient interest -> choice overload

Project Management / Agile development

Past: In TB1, we have weekly meetings and constant communication within our team and between client, to discuss workload and what to be done.

However, it can be quite chaotic sometimes to decide what to do next or not knowing priority of features. There are few times we are rushing deadline to implement features.

Now: TB2 so far we have implemented following strategies to improve the process.

- More organized (tasks to do)
- Rotating PM
- Roadmap with Sprints
- Better issues and PRs
- Kanban
- Use of AI and documentation

Project Management - Sprint 1, PM: Freddie

Roadmap for SEP TB2:

Sprint 1: MVP Release and Testing day

Week 13 - 14 (19th - 30th Jan), Project Manager: Freddie

Features planned to be finish:

- Butterfly character at item listing page
 - click buy and flying around
 - not too invasive
- Log-in page interaction
- Pop-up when something is achieved
- Basic UI for all pages in frontend
- Get ready for what to test (ethics and information form)

Fully Achieved:

- Log-in page interaction
 - has login and sign in page that links to 2 other pages
 - backend authorization has not been done
- Basic UI for all pages in frontend
 - Has marketplace, sell, explore and profile page.
- Get ready for what to test (ethics and information form)

Partially Achieved:

- Butterfly character
 - it is accessible at explore page after all items have been swiped
 - It is supposed to interact when users swipe left or right (possible idea)

Postponed / Cancelled:

- Pop-up when something is achieved
 - A bit out of stretch, did not have enough time

Now speaking: Euan

Project Management - Sprint 2, PM: Euan

Sprint 2: Beta Release & Client In-person Meeting

Week 15 - 17 (2nd - 19th Feb), Project Manager: Euan

Features planned to be finish:

- Swiping feature fully done
- Log in page with backend auth
- Connecting frontend and backend
- Liked-items page
- Dev and User instruction
- Project structure
- AI Tools folder
- Profile page simplify
- Unit tests for CI
- Compatible to android
- User flow
- Issues description change
- Connect to AWS
- Light Mode Frontend
- Research and explain why UI and gamification designs are chosen
- Delete home and merge log-in to profile page
- Beta Viva slides (rough)

Fully Achieved:

- Project structure [#103](#)
- Issues description change [#123](#)
- Dev Instruction [#86](#)
- AI Tools folder [#110](#)
- Profile page simplify [#145](#)
- Compatible to android [#125](#)
- Connect to AWS [#148](#)
- Research about UI and gamification designs [#65](#)
 - This is still an ongoing issue, it is achieved in this stage
- Delete home and merge log-in to profile page [#162](#)

Postponed / Cancelled:

- Light Mode Frontend:
 - We considered about it and about client's requirement, and found out that this is not necessary or having high priority. It has progress [#161](#) that might be continued in future
- Beta Viva slides (rough):
 - We were a bit rushing towards to the end of this sprint (client meeting) and this was ignored

Over-Achieved / Unexpected feature:

- Simplify marketplace product cards [#160](#)

Partially Achieved:

- Swiping feature:
 - Added like button, fixed background color and added butterfly interaction [#169](#)
 - This is still ongoing and new features about this will continue to be implemented
- Connecting frontend and backend:
 - Sell page is done [#148](#)
 - Other pages is not done, such as login and liked-items
- Liked-items page
 - Frontend is done [#131](#)
 - Backend is not done yet [#170](#)
- User instruction / user flow is in progress [#135](#)
- Frontend CI for unit tests in progress (no PR created yet)
- Frontend unit tests some basic tests done, more to be developed [#149](#)



Issue / PR linked
to each feature /
task

Project Management - Sprint 3 (future plan), PM: Alex

- To-do-list considered about priority and categorized in features and repos.
- Having better vision and knowledge of what to do next.
- This is discussed right after the end of sprint 2, and who to do what.
- Might try using story points for distributing workload and burndown chart to monitor weekly progress during a sprint.

Sprint 3: Beta Viva

Reading week - Week 19 (23rd Feb - 6th Mar), Project Manager: Alex

Features planned to be finish:

- High priority:
 - Features:
 - Sketch and discuss about client's idea
 - Swiping feature (more intuitive, like/dislike button)
 - Butterfly interaction
 - Repo:
 - Beta Viva slides (full)
 - Frontend CI and unit tests
 - Backend CI and unit tests
 - CD for backend
 - User flow / user instruction
- Low priority:
 - Features:
 - Consider about do we still want light theme
 - Repo:
 - Backend code formating (to pass pylint CI)
 - Documentation of meetings, roadmap and research findings
 - Github naming conventions

AI usage and Documentation

- Use of AI has been documented under ai-tools/readme.md

Example prompts would be:

- “This pop up from GitHub Action, `Error message`, explain to me what is wrong and what are the possible fixes”
- “This line of code was working before, `code`, explain in details why it does not work now”

AI tools used in production

AI tools have been used in the following ways

- Searching for frontend and backend templates
- Debugging code
- Understanding error messages and asking advice for fixing
- Other miscellaneous uses not directly related to code

AI tools used

- ChatGPT
- GitHub Copilot
- Gemini
- Powerpoint Designer AI

Now speaking: Freddie

Project Management - Kanban

Different Categories
for status tracking

The image shows a Kanban board with four columns, each representing a different stage of project management. Each column has a title, a count of items, and an estimate. The columns are:

- Long term** (1 item, Estimate: 0): Long term ongoing issue. Card: 2025-SecondhandMarketplace2 #65, [RESEARCH] - <Case study for features>, Priority: Medium, research, Size: Medium, Beta Release.
- Needs prerequisites** (2 / 3 items, Estimate: 0): Waiting for other issues. Card 1: 2025-SecondhandMarketplace2 #74, [FEATURE] - <CI - running unit tests>, Priority: Medium, Scope: Dev Tools, Size: Small, Secondhand Marketplace MVP release. Card 2: 2025-SecondhandMarketplace2 #142, [FEATURE] - Frontend: Explore page enhancement, enhancement, Priority: High, Scope: Front End, Size: Large, Beta Release. A green arrow points to the 'Beta Release' label on this card.
- Ready** (13 items, Estimate: 0): This is ready to be picked up straight away (not blocked by another task). Card 1: 2025-SecondhandMarketplace2 #138, [TEST] - Testing: Frontend unit tests, enhancement, Priority: High, Scope: Testing, Size: Medium, Beta Release. Card 2: 2025-SecondhandMarketplace2 #94, [OTHER] - Improve frontend documentation, Priority: Low, Scope: Documentation, Size: Small, Final Release. Card 3: 2025-SecondhandMarketplace2 #133, [FEATURE] - Frontend: Light mode UI.
- In progress** (5 / 5 items, Estimate: 0): This is actively being worked on. A green arrow points to the 'In progress' column header. Card 1: 2025-SecondhandMarketplace2 #129, [OTHER] - User flow / journey in main README, Priority: Medium, Scope: Documentation, Size: Small, Beta Release. Card 2: 2025-SecondhandMarketplace2 #143, [FEATURE] - Backend: authentication for login and sign in, enhancement, Priority: Medium, Scope: Back End, Size: Medium, Beta Release. Card 3: 2025-SecondhandMarketplace2 #128, [OTHER] - Beta MVP Slides.

Each card also includes a '+ Add item' button at the bottom.

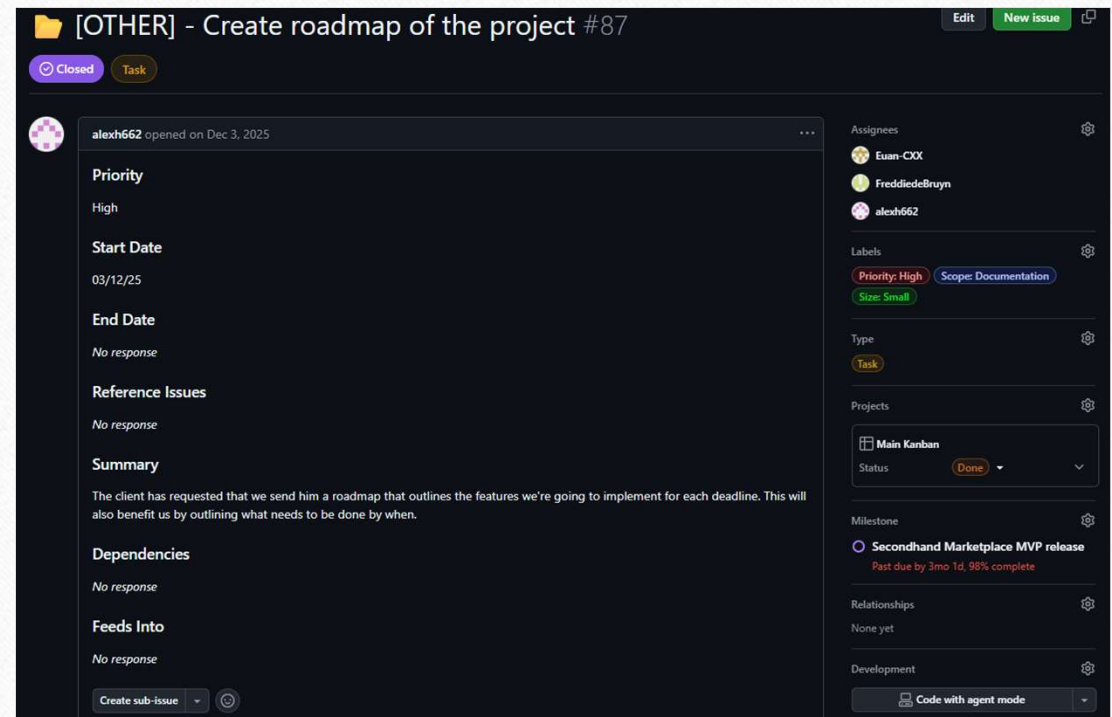
Different labels for
issues to organize

Now speaking: Freddie

Project Management - Issues before

Lacking details for others to pick up

A vague description



Project Management - Issues after

Summary has more details:

- Filepath
- Requirements clearer
- Explain issues better
- Enable other people to pick this issue up without asking for basic info

The screenshot shows a Jira issue page for a feature request. The issue title is "[FEATURE] - Connect upload item to backend and database #111". It is marked as "Closed" and "Feature". The issue was opened 3 weeks ago and edited by alexh662. The issue details include:

- Priority:** High
- Start Date:** 5/2/26
- End Date:** 8/2/26
- Summary:**
 - Implement upload item form handling found on the sell item page `frontend/app/(tabs)/sell.tsx` (line 213) by sending a POST request to the backend. The body must contain the following:
 - Title
 - Description
 - Price
 - Category
 - Images
 - Create a `backend/app/init_db.py` file to setup the database.
 - Implement POST request handling on the backend to insert the item title, description, price and category into the database as a new product, and store each image in `backend/app/static/images` with its url and reference product id stored as a new `product_image`.

The right sidebar shows the issue's metadata:

- Assignees:** alexh662
- Labels:** Priority: High, Scope: Back End, Scope: Documentation, Scope: Front End, Size: Medium
- Type:** Feature
- Projects:** Main Kanban, Status: Done
- Milestone:** Beta Release, Past due by 6d, 67% complete
- Relationships:** None yet
- Development:** Code with agent mode

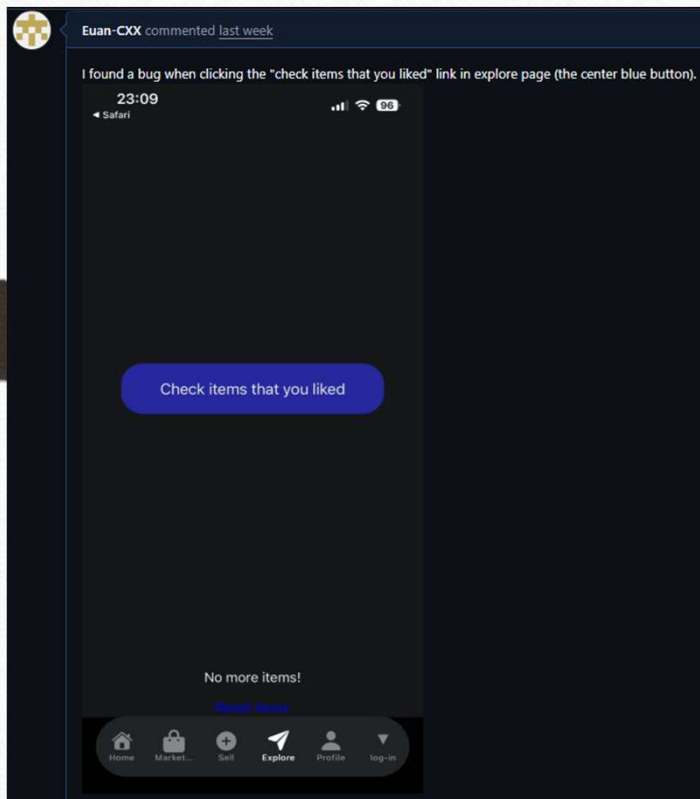
Now speaking: Alex

Project Management - PRs

☐	 6 Open	☒ 49 Closed	Author ▾	Label ▾	Projects ▾	Milestones ▾	Reviews ▾	Assignee ▾	Sort ▾
☐		Simplify the Marketplace page & add like button	enhancement	Priority: High	Scope: Front End	Size: Medium	1		16
#160 by Jen0821 was merged 3 days ago • Approved 4 of 16 tasks Beta Release									
☐		Frontend update & cleanup	Priority: Medium	Scope: Documentation	Scope: Front End	Size: Medium			11
#77 by FreddiedeBruyn was merged on Dec 3, 2025 • Approved 7 of 18 tasks Secondhand Mar...									
☐		Frontend profile page ✖	enhancement	Priority: High	Scope: Front End	Size: Medium	1		10
#100 by Euan-CXX was merged 2 weeks ago • Approved 6 of 11 tasks Secondhand Mar...									
☐		Implement liked item list frontend	enhancement	Priority: High	Scope: Front End	Size: Medium	1		10
#131 by Jen0821 was merged 2 weeks ago • Approved 4 of 18 tasks Beta Release									
☐		Project Structure	Priority: Low	Scope: Documentation	Size: Small		1		9
#107 by Jen0821 was merged 2 weeks ago • Approved 3 of 18 tasks Beta Release									
☐		Database update with ER diagram & ai-tools	Priority: Medium	Scope: Back End	Size: Small		1		9
#85 by Euan-CXX was merged on Jan 24 • Approved 8 of 14 tasks Secondhand Mar...									

Well commented
and reviewed

Project Management - PRs example comments



This is the error message.

```
ERROR [TypeError: Cannot read property 'id' of undefined]

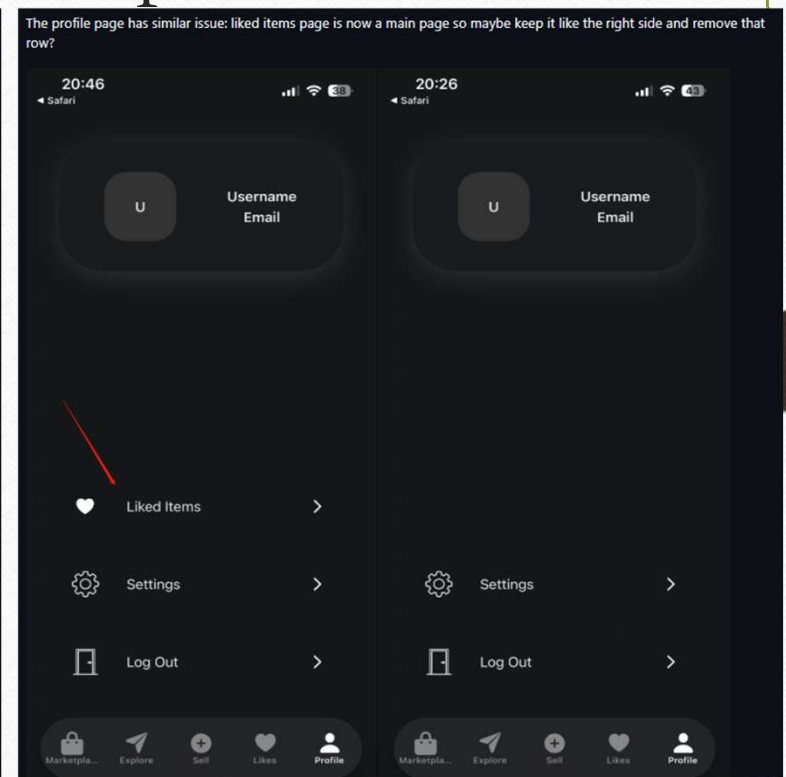
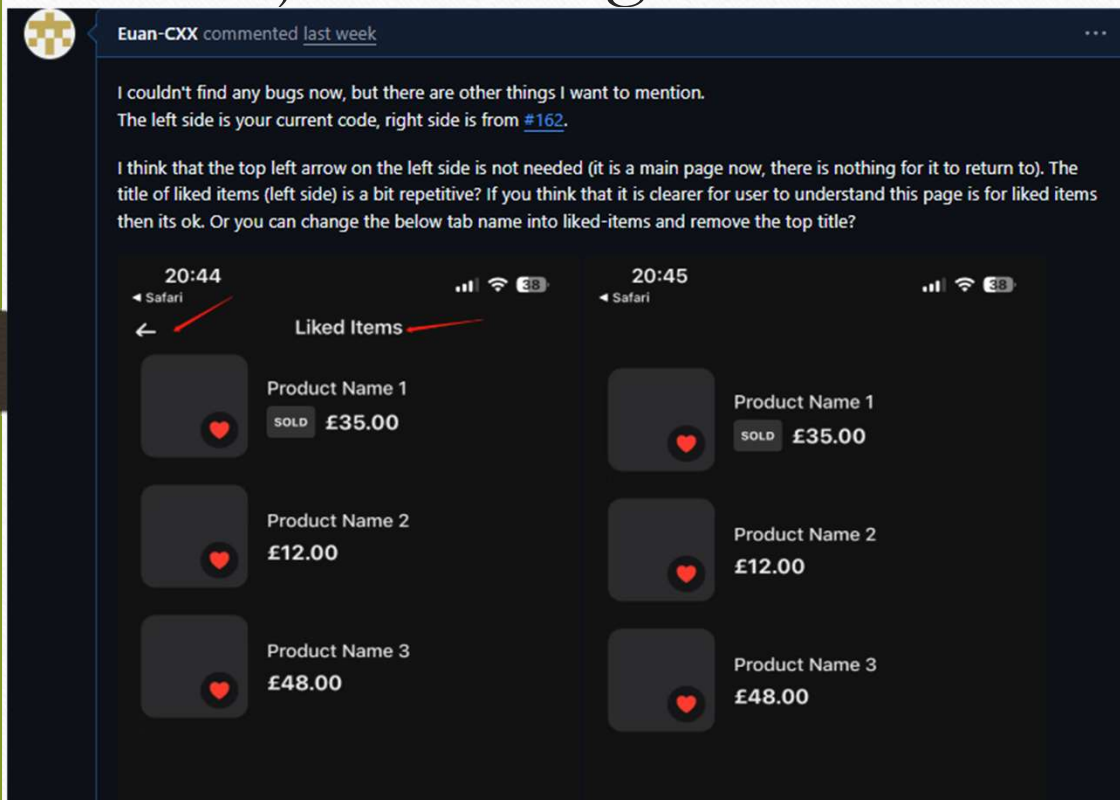
Code: [id].tsx
18 | const itemData = TestData.items[id - 1];
19 | const { toggleLike, isLiked } = useLikedItems();
> 20 | const liked = isLiked(itemData.id);
    |                                     ^
21 | // const navigation = useNavigation();
22 |
23 | const blurhash =

Call Stack
  HomeScreen (app/items/[id].tsx:20:33)

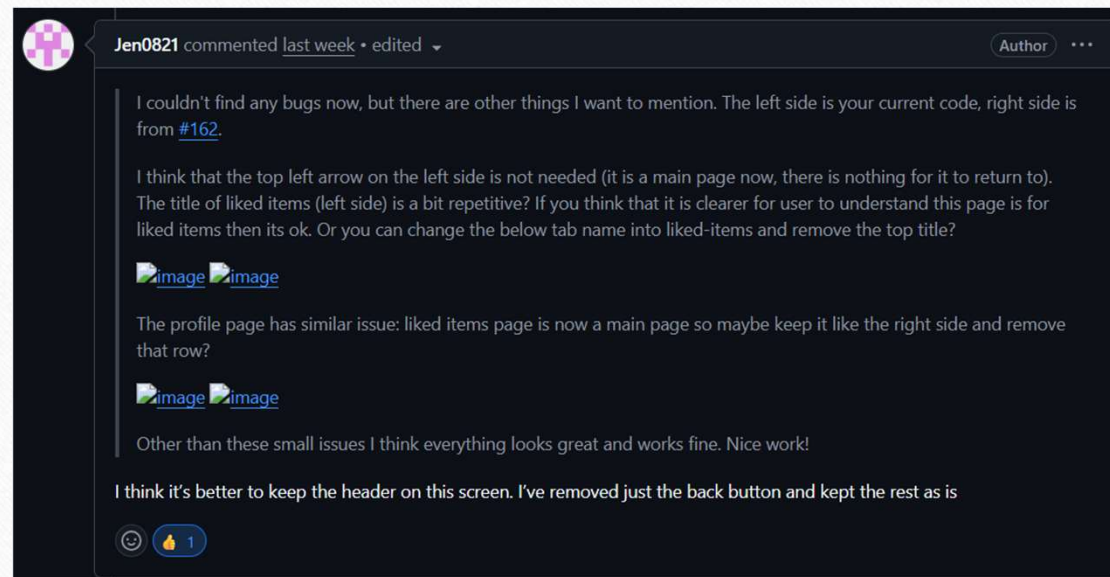
Code: _layout.tsx
3 | export default function Modallayout() {
4 |   return (
> 5 |     <Stack screenOptions={{ headerShown: false }}>
    |     ^
6 |       <Stack.Screen name="[id]" />
7 |     </Stack>
8 |   );
```

Now speaking: Alex

Project Management - PRs example comments



Project Management - PRs example comments



Challenges faced: Facing new project and new member

- New project, new framework and tech stack -> delayed start requiring more effort at a time to achieve same progress (e.g. CI facing dependency issues till recent)
- New member to integrate in Jan (TB2), working online for 2 weeks

Solution:

- Set up the new member, multiple online meetings within 2 weeks, repo and code run through, introducing with client
- Use of AI to summarize documentation and debug; break down features / ideas into smaller issues and schedule to achieve one at a time

Challenges faced: PR review stuck

- We only had 3 person in TB1 and currently 4 in TB2 in our team
- PR can get stuck / hanging for more than 1 week without enough people reviewing
- Delay future feature implementation and does not show commits in dev

Solution:

- More communication on Github, Instagram and Teams about reviewing and others
- Turn on Github Email notification
- Mention about PR review in weekly in-person meetings
- Update in group chat about PR has been created

Challenges faced: Facing client requirements

- Our current client is open to any ideas, as long as we can explain why
- Proposes idea but is up to us to decide if we want to implement it
- Letting us to explore ideas in a broad perspective (gamification) to work on
- Can be quite difficult to think of what features to actually do

Solution:

- Discuss with client about wanted effects and rough ideas
- Discuss with team about what could be done and research about effects
- Showing prototype of idea to client and discuss if it reached the requirements (e.g. swiping feature)

What we might have done differently

- Implement CI / CD earlier
- Start using sprints and roadmap earlier
- Communicate with teammates more from the beginning
- Start research earlier: about gamification, UI/UX, effect on online market etc.
- More testing for frontend and backend, and earlier

Thank you!